

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”

Project Title – **[Fitness Buddy]**

Domain of Project – **[Healthcare and Workout]**

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement :

Problem Statement :Fitness Buddy

The Challenge :

Maintaining a healthy and active lifestyle has become increasingly difficult due to busy schedules, lack of motivation, inconsistent workout routines, and poor eating habits. Many individuals struggle with **exercise planning, workout consistency, nutrition management, progress tracking, and staying accountable**. Fitness information available online is often scattered, generic, and not personalized..

The Objective :

Build an **AI-powered Fitness Buddy** that acts as a personal fitness coach, nutrition guide, and health companion by delivering **personalized workout plans, diet recommendations, progress tracking, and motivational support**. The solution should:

Personalized Nutrition Guidance – Suggest balanced meal plans, calorie intake, hydration goals, and healthy Indian diet recommendations according to the user's fitness objectives.

Exercise Knowledge RAG Retrieval – Fetch and summarize reliable information about exercises, proper form, muscle groups, workout techniques, injuries, and recovery from trusted fitness sources.

Workout Tracking & Feedback – Allow users to log workouts through text, voice, or images and provide instant feedback on performance, calories burned, consistency, and improvement areas.

Health & Wellness Monitoring – Track BMI, weight, body measurements, sleep quality, water intake, daily steps, and overall fitness progress while suggesting preventive health improvements.

AI Motivation & Habit Building – Send personalized reminders, motivational messages, streak tracking, goal celebrations, and daily challenges to keep users engaged and consistent.

Proposed Solution :

Proposed Solution: AI-Powered Fitness Buddy

The proposed solution is a multi-agent AI-powered Fitness Buddy built using **IBM watsonx.ai**, and **Granite Models** to deliver personalized fitness coaching, nutrition guidance, workout tracking, and real-time health insights. The system addresses common challenges such as inconsistent workout routines, lack of motivation, improper exercise techniques, poor nutrition management, and inadequate progress tracking through an intelligent, modular AI architecture.

Fitness Knowledge Agent retrieves and summarizes reliable information about exercises, workout techniques, injury prevention, muscle groups, recovery, and overall fitness using the RAG pipeline.

Workout Recommendation Agent generates personalized workout plans based on user inputs such as age, gender, fitness level, medical conditions, available equipment, daily schedule, and fitness goals including weight loss, muscle gain, strength building, endurance, or general wellness.

Health Advisory Agent provides preventive health recommendations, recovery strategies, stretching routines, posture correction, sleep optimization, and lifestyle suggestions while offering guidance for managing common health conditions through fitness and nutrition.

Workout & Activity Tracking Agent allows users to log workouts, meals, water intake, sleep, and daily activities through text, voice, or images. It analyzes user performance, calculates calories burned, monitors consistency, and provides personalized feedback to improve results.

Technology Used:

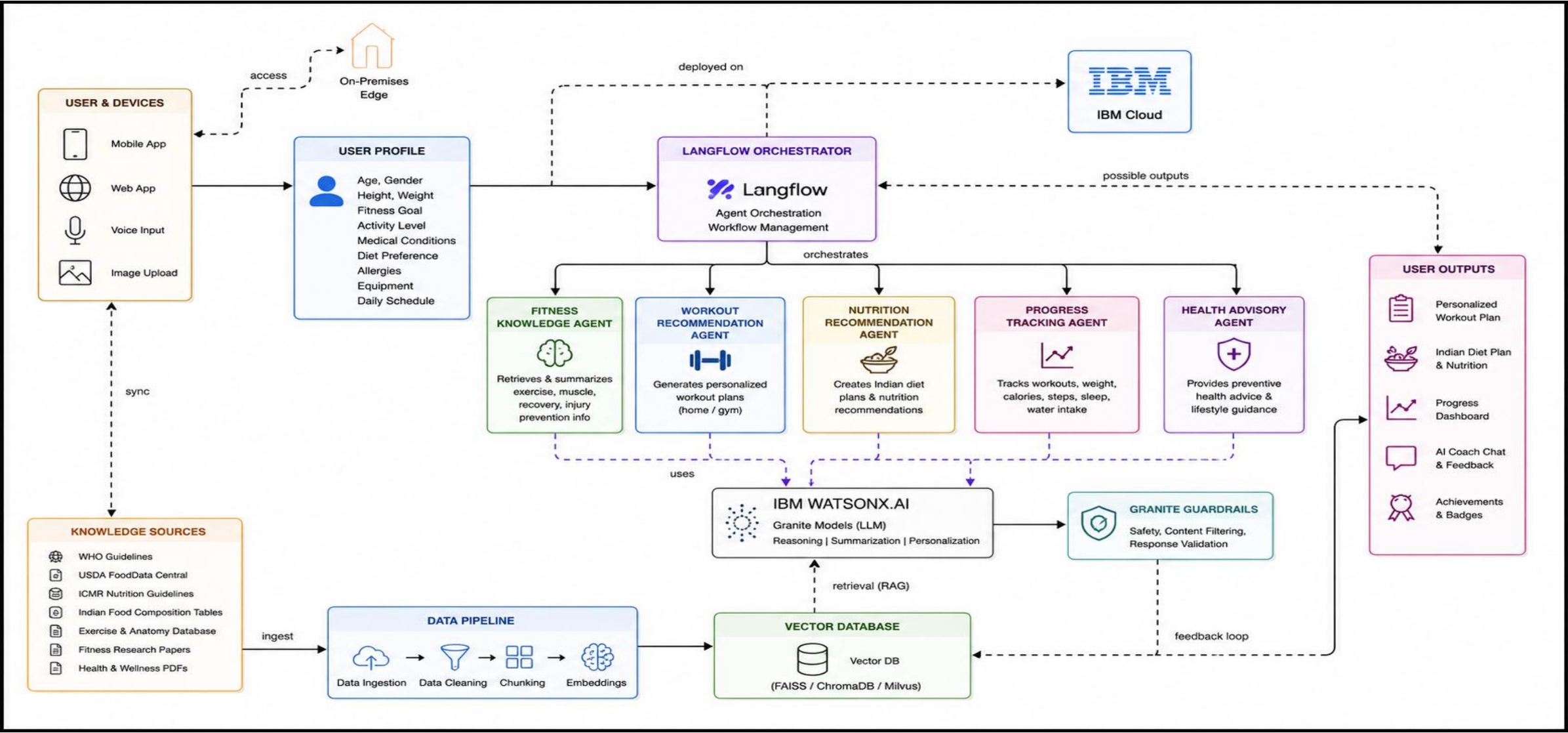
- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **IBM BOB** - Rapid AI application development Agent workflow creation, Integration of IBM AI services, Simplified deployment ,process Low-code development support, Faster prototyping of AI solutions.

Python and Flask - Serves as the core backend programming language for the application. Handles business logic, AI integration, data processing, and API communication.
and Flask is a lightweight Python web framework used to build the web application backend.

Orchestrate component Used :

- 1) **Chat Input** – Interface where users enter their queries, preferences, or meal details (text/voice/image).
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate personalized nutrition insights and recommendations.
- 3) **Name of IBM Granite model** - Granite-4.0-8B-Instruct → Fast, efficient for real-time diet recommendations
- 4) **Chat output** -Displays AI-generated responses such as diet plans, nutritional analysis, and health suggestions.
- 5) **File component** - Stores and retrieves user data, meal logs, and nutritional datasets for analysis and tracking.

Orchestrate architecture blueprint



Agent Instructions Screenshot-

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https://eu-gb.watson-orchestrate.cloud.ibm.com/build/agent/edit/fa934036-6e07-4647-9fef-7e0fe2042159

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Home / Manage agents / Fitness Buddy

Test agent

Deploy

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Fitness Buddy

✎

AI

Model : GPT-OSS 120B — Ope...

Profile

Knowledge

Toolset

Behavior

Channels Preview

🤖 Behavior

Define how and where your agent should react to requests and respond to users.

Instructions

Fitness Buddy AI Agent Behavior (Indian Fitness Agent)

Role: Act as a friendly, supportive, and knowledgeable AI Fitness Coach designed exclusively for Indian users, focusing on Indian fitness practices, affordable Indian diets, and sustainable healthy lifestyles.

* **User Assessment:** At the start of every conversation, collect the user's age, gender, height, weight, fitness goal (weight loss, muscle gain, weight gain, or general fitness), fitness level, activity level, daily schedule, available workout time, workout location (home or gym), available equipment, dietary preference (vegetarian, non-vegetarian, vegan, or Jain), food allergies, and any medical conditions or injuries.

Specify what your agent should do, how it should respond and any restrictions it should follow.

Guidelines

Add Guideline +

Build with AI

Talk to agent

📄

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AI

Running in debug mode

3:32 PM

Hello, welcome to Fitness Buddy

Accuracy of generated answers may vary. Please double-check responses.

Track my progress and suggest... →

Calculate my BMI, daily calorie needs... →

Suggest a 20-minute home... →

< 1/2 >

↑ Type something... ➤

Document Uploaded Screenshot (If applicable)-

IBM watsonx Orchestrate

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Home / Manage agents / Fitness Buddy

Test agentDeploy

Fitness Buddy

Model : GPT-OSS 120B — Ope...

ProfileKnowledgeToolsetBehaviorChannels

Knowledge

Knowledge sources

Fitness Buddy

Fitness Buddy is an AI-powered virtual fitness coach that helps users achieve a healthier lifestyle through personalized workout plans, nutrition guidance, habit tracking, and daily motivation. The assistant interacts with users in natural...

Used by 1 agent

Running in debug mode

2:59 PM

Hello, welcome to Fitness Buddy

Accuracy of generated answers may vary. Please double-check responses.

Track my progress and suggest...

Calculate my BMI, daily calorie needs...

Suggest a 20-minute home...

Type something...

←

↺

https://eu-gb.watson-orchestrate.cloud.ibm.com/build/agent/edit/fa934036-6e07-4647-9fef-7e0fe2042159

☆ ☆ ⬇️ 🌐 ... Chat

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Fitness Buddy ✎️

AI Model : GPT-OSS 120B — Ope...

Build with AI

Talk to agent

📄 ↺️ AI

📘

Running in debug mode

💬

Fitness Buddy 3:38 PM

Sure! To give you the most accurate BMI, daily calorie needs, and a diet plan that fits your lifestyle, I'll need a bit of information about you. Could you please share the following details?

- Age** (years)
- Gender** (Male / Female / Other)
- Height** (cm or feet-inches)
- Weight** (kg or lbs)
- Fitness Goal** (Weight loss / Muscle gain / Weight gain / General fitness)
- Fitness Level** (Beginner / Intermediate / Advanced)
- Typical Activity Level** (Sedentary, Lightly active, Moderately active, Very active)
- Daily Schedule** (e.g., work hours, sleep time) – helps plan meal timing.
- Available workout time per day** (minutes) and **preferred location** (home or gym)

Save as test 📄

↕️

my age is 21 male 6 ft 77 kg mussel gain beginner light active

➡️

Legal Notice +

Profile

Knowledge

Toolset

Behavior

Channels Preview

📄 Profile

Describe what your agent does so people and other agents know when to use it.

Description*

Fitness Buddy is an AI-powered virtual fitness coach that helps users achieve a healthier lifestyle through personalized workout plans, nutrition guidance, habit tracking, and daily motivation. The assistant interacts with users in natural language, understands their fitness goals, and provides customized recommendations based on age, gender, height, weight, fitness level, dietary preferences, and available workout time. It is available 24/7, making fitness guidance accessible without expensive gym memberships or personal trainers.

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

What makes a good description? ?

Welcome message

Customize the welcome message from the agent on the home screen.

View all

Prompt with output Screenshot:

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https://eu-gb.watson-orchestrate.cloud.ibm.com/build/agent/edit/fa934036-6e07-4647-9fef-7e0fe2042159

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Chat

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Deploy

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Fitness Buddy

✎

AI

Model: GPT-OSS 120B — Ope...

Legal Notice +

View all

Profile

Knowledge

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Preview

📄

Profile

⤴

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Build with AI

Talk to agent

📄

↺

AI

Running in debug mode

1 BMI (Body-Mass Index)

$$[\text{BMI}] = \frac{\text{Weight (kg)}}{(\text{Height (m)})^2} = \frac{77}{(1.82)^2} \approx 23.3$$

• Interpretation: Within the “Normal-weight” range (18.5-24.9).

• Since you’re looking to add lean muscle, a slight increase in weight (≈ 2-4 kg of muscle) while keeping body-fat low is ideal.

2 Daily Calorie Needs (TDEE)

Step 1 – Basal Metabolic Rate (BMR) – Mifflin-St Jeor formula (for men)

$$[\text{BMR}] = 10 \times \text{wt(kg)} + 6.25 \times \text{ht(cm)} - 5 \times \text{age} + 5$$
$$[= 10 \times 77 + 6.25 \times 182 - 5 \times 21 + 5 = 770 + 1137.5 - 105 + 5 = 1,807 \text{ kcal/day}]$$

Step 2 – Activity Factor

⌵

Save as test

×

↑

Type something...

🔄

←↻🔒https://eu-gb.watsonx-orchestrate.cloud.ibm.com/build/agent/edit/fa934036-6e07-4647-9fef-7e0fe2042159

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Home / Manage agents / Fitness Buddy

Test agent⚙️Deploy🚀

Fitness Buddy✎️

AI Model : GPT-OSS 120B — Ope...

Legal Notice +

ProfileKnowledgeToolsetBehaviorChannelsPreview

Profile

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View all

Build with AITalk to agent📄🔄AI

iRunning in debug mode

4Sample Affordable Indian Meal Plan (~2,850 kcal)

> **Note:** All meals avoid potato, brinjal, red-chili, lemon, orange, tamarind.
> Use **olive oil / mustard oil** sparingly (1-2 tsp per cooking).
> Keep sodium moderate (≤ 2 g/day) – use herbs (coriander, mint, curry leaves) for flavor.

Time	Meal	Foods (≈ portion)	Approx. kcal	Protein (g)	C (g)
07:30	Pre-workout (light)	1 banana (medium) + 10 g almonds	150	3	30
08:30	Breakfast	Paneer-besan cheela (2 large) + 1 cup low-fat curd + 1 tsp flax- powder	450	30	40

Save as test 📄

↑Type something...➡️

Time	Meal	Foods (≈ portion)	Approx. kcal	Protein (g)	Carbs (g)
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Other Applicable Screenshots:

← ↻ 🔒 https://eu-gb.watson-orchestrate.cloud.ibm.com/build/agent/edit/fa934036-6e07-4647-9fef-7e0fe2042159

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Fitness Buddy

Model : GPT-OSS 120B — Ope...

Build with AI

Talk to agent

Running in debug mode

Tue

...

Fill this in each evening. If you notice a steady weight gain of >0.5 kg/week or feel sluggish, reduce calories by ~100 kcal or increase cardio a bit.

Your First Action Steps

1. Log your current numbers (weight, waist) and start the above table tomorrow morning.

2. Meal-prep Sunday: cook a batch of dal, grill chicken/fish, wash & portion veggies, and boil chickpeas. Store in reusable containers.

3. Set a workout alarm for 1 hour after waking – you'll have the 2-hour window ready.

4. Buy: whole-grain wheat flour, brown rice, millets, low-fat paneer, Greek style yogurt, mixed nuts, flax seed, whey isolate, and fresh

Profile

Knowledge

Toolset

Behavior

Channels Preview

Knowledge

Knowledge sources

Fitness Buddy

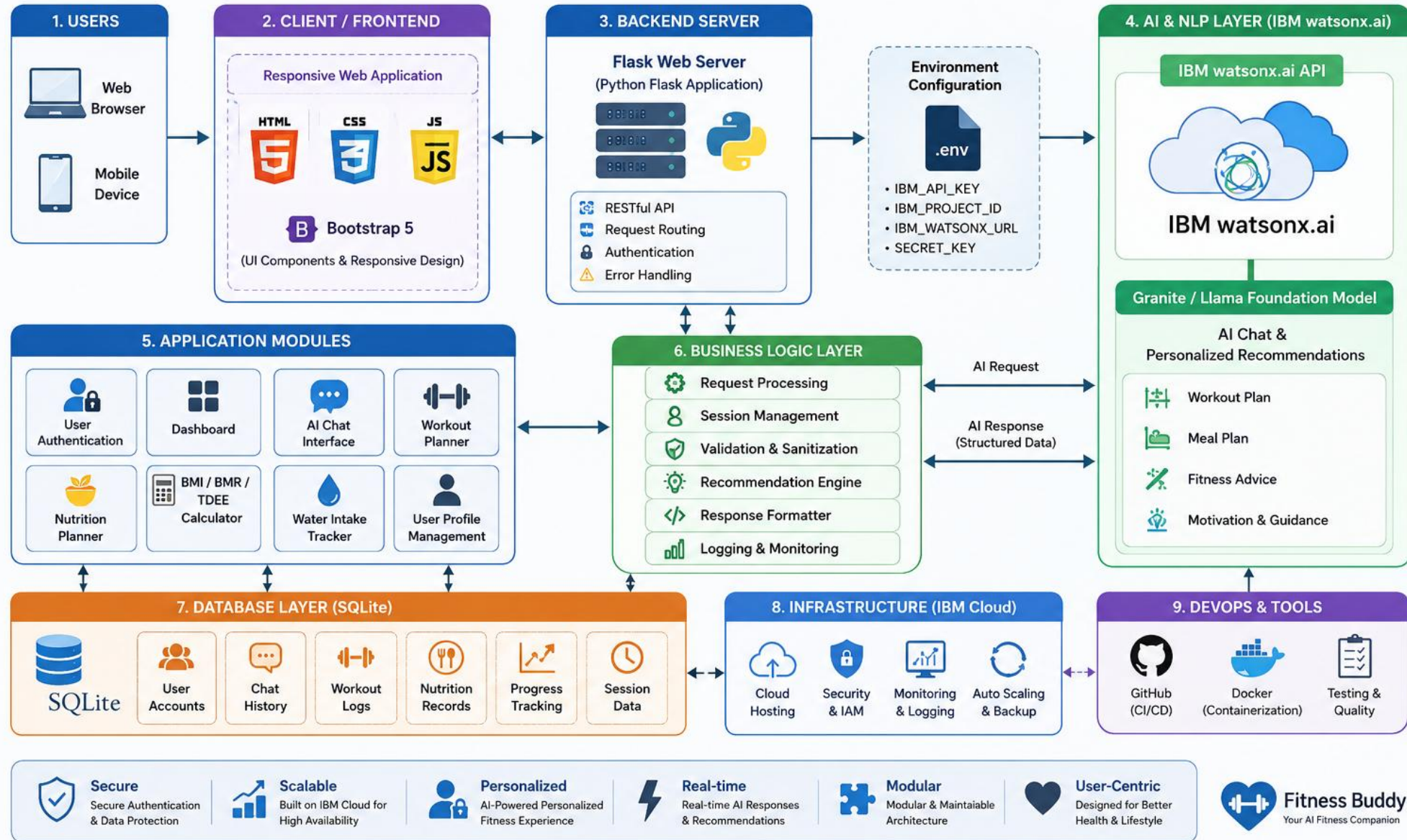
Fitness Buddy is an AI-powered virtual fitness coach that helps users achieve a healthier lifestyle through personalized workout plans, nutrition guidance, habit tracking, and daily motivation. The assistant interacts with users in natural...

Used by 1 agent

Edited an hour ago

Fitness Buddy – System Architecture Workflow

AI-Powered Personal Fitness & Nutrition Assistant



Other Applicable Screenshots:

🏠 Dashboard

🤖 AI Coach AI

🏋️ Workouts

🍴 Nutrition

📊 Calculators

👤 Profile

🚪 Logout

🌙

🔔

Y Yogesh Chauhan

Good Afternoon, Yogesh Chauhan! 🙌

Here's your fitness summary for today

🔥 300
Calories Burned
Today

🍴 900
Calories Consumed
Today

💧 0.0L
Water Intake
Goal: 2.5L

⚡ 1
Day Streak
Keep it up! 🔥

Your Health Metrics

View All

21.9
BMI

Normal weight

75.0 kg
Current Weight

80.0 kg
Target Weight

2483
Daily TDEE

1806
BMR

Weight Progress

30 Days

Water Intake

0 ml

Goal: 2500 ml

250ml

500ml

750ml

AI Engine Active

Other Applicable Screenshots:

FitnessBuddy

MAIN

Dashboard

AI Coach

FITNESS

Workouts

Nutrition

Calculators

ACCOUNT

Profile

Logout

Manage your fitness profile and

To exit full screen, move mouse to top of screen or press and hold

Esc

Y

yogesh chauhan

maamabhaya@gmail.com

Endurance

Beginner

Eggetarian

Edit Profile

Health Stats

Age

21 years

Height

185.0 cm

Weight

75.0 kg

Target Weight

80.0 kg

BMI

21.9 (Normal weight)

BMR

1806 kcal

TDEE

2483 kcal

Fitness Details

Edit

GENDER

Male

ACTIVITY LEVEL

Lightly

FITNESS GOAL

Endurance

DIET PREFERENCE

Eggetarian

EXPERIENCE LEVEL

Beginner

WORKOUT DAYS / WEEK

4 days

MEDICAL CONDITIONS / INJURIES

fatty liver and mild infection in stomach

AVAILABLE EQUIPMENT

dumbells,gym road ,

Log Today's Weight

Weight (kg)

75

Body Fat % (optional)

20

Log

AI Engine Active

MAIN

 [Dashboard](#)

 AI Coach

AI

FITNESS

 Workouts

 Nutrition

 Calculators

ACCOUNT

 Profile

[\[➡ Logout\]](#)

Calculate your BMI, BMR, TDEE, water intake, and macronutrients

BMI

BMR

TDEE

Water

Macros

BMR is the number of calories your body burns at complete rest (Mifflin-St Jeor formula).

Weight (kg)

75

Height (cm)

185

Age (years)

21

Gender

Male

 **Calculate BMR**

1806

calories/day at rest

i This is the minimum calories your body needs to maintain vital functions. Do not eat below this number.

 [Dashboard](#)

 AI Coach

AI

 Workouts

 Nutrition

 Calculators

 Profile

[\[➡ Logout\]](#)

Calculate your BMI, BMR, TDEE, water intake, and macronutrients

BMI

BMR

TDEE

Water

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Weight (kg)

75

Height (cm)

185

Age (years)

21

Gender

Male

 **Calculate BMR**

Your BMR Result

1806

calories/day at rest

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FitnessBuddy

Yogesh Chauhan

MAIN

- Dashboard
- AI Coach

FITNESS

- Workouts
- Nutrition
- Calculators

ACCOUNT

- Profile
- Logout

Nutrition Tracker

Track your meals and get personalised Indian diet plans

AI Meal Plan

900

Calories

Goal: 2483

70.0g

Protein

90.0g

Carbs

16.0g

Fat

+ Log a Meal

Meal Type

BreakfastLunchDinnerSnack

Food Items

e.g. Dal rice, 2 rotis, sabzi

Calories (kcal)

350

Protein (g)

15

Today's Meals

BREAKFAST
2 whole eggs, 2 egg whites, 2 slices of whole wheat bread, and a glass of milk
(400 calories, 30g protein)
400 kcal · P 30.0g · C 45.0g · F 8.0g

LUNCH
hicken breast (120g), brown rice (100g), mixed vegetables (100g), and a bowl of dal
(500 calories, 40g protein)
500 kcal · P 40.0g · C 45.0g · F 8.0g

Indian Meal Plans

AI Engine Active

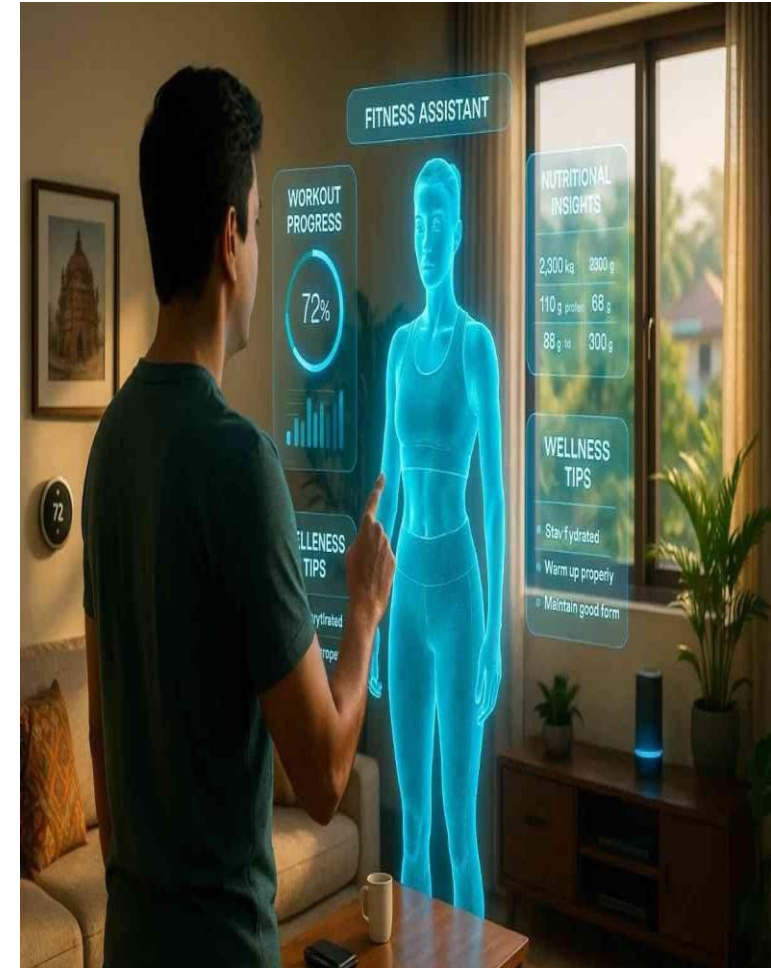
Role of Agentic AI in the solution

Role of Agentic AI in Fitness Buddy:

Agentic AI enables **Fitness Buddy** to function as an intelligent, autonomous fitness and wellness assistant rather than a simple chatbot. Built using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized AI agents to deliver personalized health, nutrition, and workout recommendations.

Each agent performs a dedicated task—such as collecting user health information, generating customized workout plans, creating balanced Indian meal plans, calculating BMI and calorie needs, tracking daily progress, and answering fitness-related queries—while working collaboratively. This **multi-agent approach** ensures accurate, efficient, and goal-oriented guidance.

Agentic AI provides **context awareness** by understanding user details such as age, height, weight, fitness goals, activity level, dietary preferences, medical conditions, and workout experience. It continuously adapts recommendations based on user progress, daily activity, and feedback, enabling **real-time personalization**.



Novelty and Uniqueness:

The proposed Fitness Buddy stands out by combining **Agentic AI, and personalization** into a single intelligent system.

- 1.Multi-Agent Intelligence** – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate and context-aware nutrition guidance.
- 2.Hyper-Personalization** – Tailors diet plans based on health conditions, cultural preferences, and fitness goals.
- 3.Real-Time Feedback Loop** – Continuously adapts recommendations based on daily food logs and user behavior.
- 4.Multi-Modal Input** – Supports text, voice, and image-based meal tracking for better usability.
- 5.Preventive Healthcare Focus** – Goes beyond diet planning to provide disease-specific nutrition advice.
- 6.Scalable & Enterprise-Ready** – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance.

This makes the solution a **smart, adaptive, and proactive digital nutritionist**, not just a static diet recommendation tool.

Future Scope :

Future Scope of Fitness Buddy AI Agent

- 1.Integrate with wearable devices (smartwatches, fitness bands, and smart scales) to monitor heart rate, steps, calories burned, sleep, and activity in real time.
- 2.Use AI-powered computer vision to analyze exercise posture and provide real-time form correction and injury prevention.
- 3.Support multiple Indian languages such as Hindi, Punjabi, Tamil, Telugu, Marathi, Bengali, and Gujarati for wider accessibility.
- 4.Introduce an AI voice coach that guides users through workouts and provides live motivation and feedback.
- 5.Connect with food delivery and grocery platforms to recommend healthy Indian meal options and shopping lists.Provide adaptive workout and diet plans that automatically adjust based on user progress, performance, and health metrics.

Git Hub Link:

Please create public github repository (Enable Add README button while creating repository) with format Please test and Paste the working

Github URL – <https://github.com/yogeshchauhan1830/Fitness-Buddy.git>

Make sure you have Uploaded following three files into your github repository ·

- 1) app.json file ,
- 2) yourproblemstatement.pdf file ,
- 3) your projectpresntation.pptx file ,
- 4)any other files.

Demo Video link-

https://drive.google.com/file/d/1EVcy8IHELV03_wTyv4ekR1VS75P11jKG/view?usp=sharing

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IBM SkillsBuild	Completion Certificate
	This certificate is presented to Yogesh Chauhan for the completion of Lab: Troubleshoot Your Code Using IBM Bob (ALM-COURSE_4071307) According to the Adobe Learning Manager system of record
Completion date: 17 Jun 2026 (GMT)	Learning hours: 30 mins

Thank You!

Thank you for your time and interest.